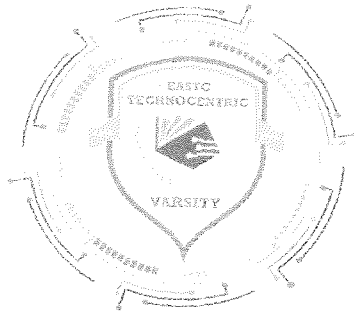


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 03	Electricity and Electronics, NQF Level 4.	13



ELECTRICITY AND ELECTRONICS
SUMMATIVE ASSESMENTS

55
60
927

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Contact Address	32467 Mondla Str Ex 12
Telephone (H)	0747732317
Telephone (W)	
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Cellular	0747732317
E-mail	collenx14@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

<u>Xolani Malinga</u>

Learners Name: XOLANI COLLEN MALINGA

Learner's signature:  Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

9/10

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>XOLANI MALINGA</i>	Signature 
Assessors Name:	Signature:

Y.C.

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1	List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications.	10

Examples of Conducting Materials

Copper

Aluminum

Silver

Gold

Iron/Steel

Electrical Properties Very low electrical resistance

Allow electric current to flow easily

Have many free electrons for conduction

Mechanical Properties

High tensile strength

Ductile (Can be drawn in wires)

good thermal conductivity

Durable and flexible

Applications

Electrical wiring and cables

Transmission lines

Motor windings

Electrical contacts and connectors

Insulators

Examples of Insulating Materials

Rubber

Plastic (PVC)

Glass

Porcelain

Mica

Electrical Properties

Very high electrical resistance

Do not allow current to flow easily

Prevent leakage of electricity

Mechanical properties

Strong and rigid

Heat resistant

good mechanical strength

Resistant to moisture and chemical

Applications

Cable insulation

Overhead line insulators

Switchgear insulation

Transformer insulation

Semiconductors

Silicon, germanium and gallium arsenide

Electrical Properties

Moderate conductivity (between conductor and insulator)

Conductivity increases with temperature

Can be controlled by doping

Task	Questions Description	Marks
2	a) Name, describe and explain the different types of magnets and their properties b) Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density	10

These are magnets that maintain their magnetic properties for a long period once magnetized. They do not lose their magnetism easily.

They are made from hard ferromagnetic materials like steel, alnico (aluminum, nickel, cobalt) and neodymium.

They have a persistent magnetic field and always have a North and South pole.

Temporary Magnets: Their magnetic strength is dependent on the external field. They are easily magnetized and demagnetized.

Electromagnets: Their strength can be adjusted by changing the current or the number of coils. The magnetism can be turned on and off instantly.

Fundamental Magnetic Concepts

Direction: Outside the magnet, they travel from the north pole to the south pole.

Inside the magnet, they travel from south to north.

Continuous loops: They form complete closed-ended loops.

No Intersection Magnetic line of force never cross or intersect each other

Strength and density the lines are most concentrated at the poles, where the magnetic field is strongest.

Tension they behave like stretched elastic bands, trying to shorten their length (which explains why opposite poles attract).

Relationship between Flux and Flux density

Magnetic Flux (Φ) this is the total number of magnetic field lines passing through a specific area.

Magnetic Flux Density (B) this refers to the thickness or concentration of the flux. It is the amount of flux passing through a unit area perpendicular to the field. It is measured in Tesla (T)

The mathematical relationship
the relationship is expressed by the formula

$$B = \frac{\Phi}{A}$$

Where:

B = Flux density (Tesla)

Φ = Total Magnetic Flux (Webers)

A = Cross-Sectional Area (m^2)

If you have a constant amount of flux, decreasing the area (A) will increase the flux density (B), making the magnetic field stronger in that specific spot. ✓

Task	Questions Description	Marks
3	Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor.	10

The right-hand grip rule is used to determine the direction of the magnetic field around a current carrying conductor.

Hold the conductor with your right hand
the thumb points in the direction of current flow (I)
the curved fingers show the direction of the magnetic field lines

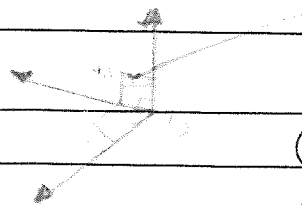
b. Around the conductor
2) Magnetic field around a conductor.

When electric current flows through a straight conductor it produces a magnetic field around it.

Characteristics
The magnetic field forms concentric circles around the wire.

The field becomes stronger when the current increases.

The field becomes stronger when current increases.
The field strength decreases as distance from the conductor increases.



Magnetic Field Strength (B) increases
Current (I) increases
Length of Conductor (L) increases

LEARNING UNIT 2:

KM-03-KT03: Earthing and bonding (35%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Define earthing and bonding and explain the purpose thereof b) Explain the regulatory requirements for earthing and bonding	10

Earthing (grounding) This is the process of connecting the non-current carrying metallic parts of an electrical installation (such as the metal casing of an appliance) directly to the earth (ground). This is typically achieved using an earth electrode buried in the soil.

Bonding (Equipotential bonding) This involves electrically connecting all exposed metalwork and extraneous conductive parts (like metal water pipes, gas pipes or structural steel) together. The goal is to ensure all these parts are at the same electrical potential.

Regulatory requirements for earthing and bonding.

Regulatory requirements ensure that electrical installations meet minimum safety standards to protect lives and property. While specific codes vary by region (e.g. SANS 10142 in South Africa, BS 7671 in the UK).

Mandatory Installation All installations must have a verified earthing system. It is illegal and unsafe to operate a mains-powered system without one.

Continuity and Resistance regulations specify maximum allowable resistance levels for the earth path. The resistance must be low enough to ensure that the fault current is high enough to trip the circuit breaker instantly.

Main Equipotential bonding All incoming metallic services (water, gas, oil) must be bonded to the main earthing terminal (MET) near the point of entry into the building.

Supplementary bonding In high-risk areas like bathrooms or swimming pools, additional bonding is often required between all accessible metal parts to minimize shock risk in wet conditions.

Labelling and Identification Earth conductors must be clearly identified (usually with green/yellow insulation) and earth terminals must be labeled to prevent accidental disconnection.

Task	Questions Description	Marks
5	a) Describe the provision of earthing for underground cables and overhead lines b) Explain the various processes of measuring, testing and calculating earthing and bonding system values	10

Earthing is a critical safety measure used to provide a low-resistance path for fault currents to reach the ground, protecting both personnel and equipment.

The primary method for earthing underground cables involves using the cable's metallic sheath (usually lead or aluminium) or its steel wire tape armour. These metallic layers are bonded to the main earthing terminal at both ends of the cable run.

Before installation, the ground's ability to conduct electricity is measured using the Wenner four-pin method. This helps determine how many electrodes are needed and how deep they must be driven.

Calculating Earth Fault Loop Impedance (Z_s)

$$Z_s = Z_e + (R_1 + R_2)$$

Z_e = External earth loop impedance (from the source)

R_1 = Resistance of the phase conductor

R_2 = Resistance of the protective earth conductor

In high-voltage substations, engineers calculate the potential difference a person might experience between their feet.

LEARNING UNIT 3:

KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
6	a) Describe the various inductors used and typical applications b) Describe the construction and uses of various semiconductor devices c) Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols	10

Air Core Inductors Used in high-frequency applications like RF (Radio Frequency) applications tuning and radio transmitters because they lack core losses

Iron Core Inductors typically used in low frequency applications such as power supplies and filters due to their high inductance.

Ferrite Core Inductors Common in high frequency applications like switching power supplies) because ferrite has high permeability and low electrical conductivity reducing energy loss

Variable Inductors Used in tuning circuits where the inductance needs to be adjusted manually.

Typical Applications

Filtering - Removing noise from signals or power lines.
Energy Storage - Used in Switch-Mode power supplies (SMPS)
Tuning - Selecting specific frequencies in radio and TV receivers.

Describe the construction and Uses of various semiconductor devices

Semiconductors are materials (like Silicon or germanium) whose electrical conductivity falls between a conductor and an insulator.

Diodes Construction formed by joining a P-type and an N-type Semiconductor (P-N junction).

Transistors (e.g. BJT, MOSFET) Construction three-layer sandwiches (NPN or PNP) or field-effect structures. Uses Act as switcher (in digital logic) or amplifiers (in audio/radio equipment).

Variable inductor

Inductance can be adjusted

Used in tuning circuits such as radios and transmitters

Diodes Construction A diode is made from two layers of ~~sm~~ semi-conductor.

Material (P-type and N-type) joined together to form a PN Junction.

Light Emitting diode (LED)

Construction An LED is a PN junction diode that emits light when current flows through it.

Uses Indicator lights, display panels, lighting systems

Integrated Circuits (IC) Construction An IC is a small

22 | Page semiconductor chip containing

Many electronic components like transistor, resistors and capacitors
Uses = Computers and smartphones, control systems, digital electronics

c)

Simple battery and lamp circuit

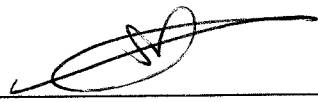
Battery, Switch and lamp circuit

Diode Rectifier Circuit

Assessor's Report

Candidate's Name	XOLANI MALIN GA	ID No.	9511155424085
Assessor's Name		Reg. No.	
Title			
ASSESSMENT DECISION			
Outcome	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
Signature of Assessor	Signature of Candidate		

Candidate Feedback Report

Candidate's Name	Xolani Collen Malinga		ID No.	9511055424085
Assessor's Name			Reg. No.	
Title				
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

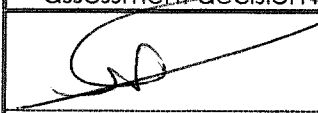
Candidate Appeal Form

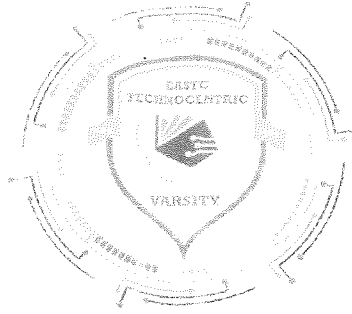
Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name		ID No.	
Title			
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER		NAME of ASSESSOR	
VENUE		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
			
Learner	Date	Assessor	Date
		Moderator	Date



40
40
100%

ELECTRICITY AND ELECTRONICS

FORMATIVE ASSESSMENT



Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

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Cellular	0747732317
E-mail	collenx14@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

Potential difference is the difference in electrical energy between two points in a circuit

Symbol: V

Unit: Volt V

Formula

$$V = \frac{W}{Q}$$

V = Voltage

Q = charge

W = Work done

Resistance is the opposition to the flow of electric current in a conductor

Symbol: R

Units: Ohm (Ω)

Factors Influencing resistance: Length of the conductor, Cross-sectional area, Material type and temperature.

Ohm's law: The current through a conductor is directly proportional to the voltage across it, provided $T^\circ C$ remains constant

Kirchhoff's laws

The total current entering a junction equals the total current leaving the junction.

$$I_1 + I_2 = I_3 + I_4$$

Kirchhoff's Voltage law

The sum of all voltage in a closed loop equals zero.

$$V_1 + V_2 + V_3 = V_{\text{supply}}$$

Series and Parallel Circuits

Current is the same in all components

Total resistance

$$R_t = R_1 + R_2 + R_3$$

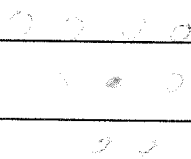
Parallel Circuit Voltage is the same across each branch

Task	Questions Description	Marks
2	a) Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated. b) Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place	10

Magnet field pattern
the magnetic field forms Circular lines around the wire
the direction of the field depends on the direction of the Current

Right-hand grip rule - Thumb - direction of Current
Curved fingers - direction of magnetic field

Current coming out of the page (o)



Current into the page (x)



Effect when placed in a uniform magnetic field
When a current carrying conductor is placed in a magnetic field, the conductor experiences a force and may

Fleming's left-hand rule
First finger (index) - direction of magnetic field
Thumb - direction of motion / force
Middle finger - direction of Current

b. Magnetic fields can be changed in several ways

Change the current

Increasing the electric current increases the strength of the magnetic field.

Change: field becomes stronger and more concentrated

Insert an Iron Core

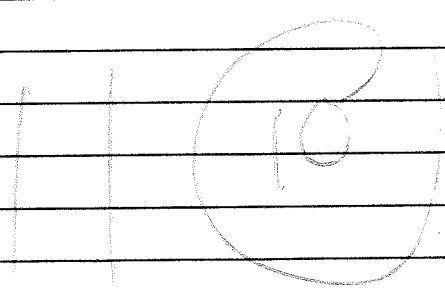
Placing a soft iron core inside a coil increases the magnetic field strength

Change: Magnetic field becomes much stronger

4. Change the distance

Magnetic field strength decreases as you move away from the conductor or magnet.

Change: field becomes weaker with distance.



LEARNING UNIT 2:

KM-03-KT03: Earthing and bonding (35%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
3	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation b) Explain the term 'systems earthing' c) Describe the earthing systems relevant to SANS 10142-1. d) Describe how a common earth electrode is used in reticulation circuits	10

the neutral conductor must be earthed at the source of supply (transformer star point or generator)

Multiple earthing points may be used along the distribution network to stabilize the system

the supplier ensures the neutral is connected to earth to limit voltage rise and provide a return path for fault current

the consumer installation must connect the neutral to the earth through the main earthing terminal (MET) only where permitted by the supply system

Main earthing point Only

The installation must have a main earthing terminal connected to an earth electrode.

proper earthing ensures fault currents flow safely to earth and trip protection devices

Systems earthing Refers to the earthing of the electrical power system usually the neutral point of a transformer or generator.

Purpose of Systems earthing

limits Voltage during faults

Provides a reference point for the electrical system allows protective devices to operate quickly

Impr

TH

Neutral is earthed at the transformer protective earth is connected to the supply earth. Common in many installations

TT System

Supply neutral is earthed by supplier Consumer has their own earth electrode Used in some rural or special installations

IT System

Supply is not directly earthed or connected through Imp

Used in timing circuits, audio circuits and signal processing

Variable Capacitors

Capacitance can be increased
Used in radio tuning circuits

Capacitors

Small but high capacitance
Used in portable electronic devices mobile phones and computers

b) Capacitors In Series

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad C_1 = 6 \mu F, C_2 = 3 \mu F$$

$$\frac{1}{C_T} = \frac{1}{6} + \frac{1}{3}$$

$$\frac{1}{C_T} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6} \quad C_T = 2 \mu F$$

Rule: Total Capacitance decreases in Series
Capacitors In parallel

$$C_T = C_1 + C_2 + C_3$$

$$C_1 = 6 \mu F, C_2 = 3 \mu F$$

$$C_T = 6 + 3 = 9 \mu F$$

Rule total Capacitance increases

In parallel

c) Brown - Black - Red - Gold

Colour Value

Brown	1
Black	0
Red	$\times 100$
Gold	$\pm 5\%$

$$10 \times 100 = 1000 \Omega$$
$$1k\Omega \pm 5\%$$

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

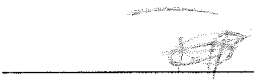
1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1				
2				
3				
4				



Learner Signature

Date



Facilitators' Signature

Date

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

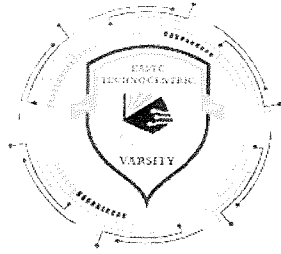
What resources would you require to conduct the assessment?

1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit

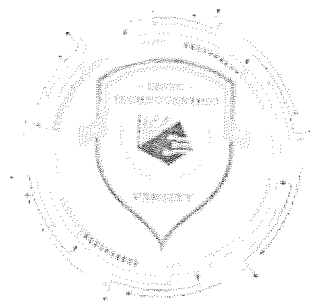


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 04	Industry Context, NQF Level 3.	2



25
30
832

INDUSTRY CONTEXT

SUMMATIVE ASSESMENT



Module #	671101000-KM-04:
NQF Level	level 3,
Notional hours	20
Credit(s)	Cr 2
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Contact Address	32467 Mandela Str Gx12
Telephone (H)	0747732317
Telephone (W)	0747782317
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

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Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

<u>Xolani Malinga</u>

Learners Name: XOLANI COLLEN MALINGA

Learner's signature:  Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____



PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Xolani Malinga</i>	Signature: 
Assessors Name:	Signature:

Xolani Malinga

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-04-KT01: Introduction to the world of work and the electrical trade (100%)

Learning Outcome 1:	KT0101 The Electricians world of work
Learning Outcome 2:	KT0102 Career opportunities of qualified Electricians
Learning Outcome 3:	KT0103 The Electrician's responsibilities and duties
Learning Outcome 4:	KT0104 Legislation relating to apprentices in the Electrical trade
Learning Outcome 5:	KT0105 Trade test requirements

Task	Questions Description	Marks
1	Describe and explain the job environment and workplace roles of an Electrician are accurately	10

An electrician works in different environments depending on the type of work they do. They may work in homes, commercials buildings, factories, construction sites and outdoor locations.

The job environment can sometimes be dangerous because electricians work with electricity tools and sometimes at heights or in confined spaces. Because of this electricians must follow strict safety rules and wear protective equipment such as gloves, safety boots, helmets and protective clothing.

Electricians often work both independently and as part of a team with other professionals like engineers, builders and technicians. They may work in different weather conditions when doing outdoor electrical installations or repairs.

The workplace roles of an electrician include installing, maintaining, testing and repairing electrical systems and equipment.



Task	Questions Description	Marks
2	Describe the profile of an Electrician with respect to industry descriptions, career path progression and requirements	10

An electrician is a skilled tradesperson who installs, maintains and repairs electrical systems in homes, buildings, factories and other facilities. Electricians ensure that electrical equipment and wiring operate safely and efficiently according to industry standards and safety regulations. Electricians work in different industries such as construction, manufacturing, mining, maintenance and energy sectors. Their duties include installing electrical wiring, connecting circuits, testing electrical systems, repairing faults and maintaining equipment. They use tools like multimeters, pliers, screwdrivers and electrical testing equipment. Safety is very important in this profession because electricians work with high-voltage electricity.

Requirements to become an electrician need a high school certificate, especially Mathematics and physical Science and technical training at a TAFE college or training institute. Completion of an apprenticeship program, passing a trade test to become a certified electrician.

9

Task	Questions Description	Marks
3	Explain Trade test methodology, requirements and assessment procedure leading up to the trade assessment	10

The trade test methodology refers to the structured process used to evaluate whether a learner has the knowledge and practical skills required to perform competently in a specific trade. The process ensures that the candidate meets the national occupational standards.

The candidate must have completed the required training or apprenticeship in the relevant trade. A logbook or workplace evidence of practical experience must be completed and signed. The learner must meet the minimum practical training hours required by the training authority. The candidate must apply for the trade test at an accredited trade centre.

All necessary documents such as ID, training Certificates and logbook must be submitted. Pre-assessment/Screening the training centre checks the learners documents and readiness for the trade test.

Knowledge Assessment the candidate may complete a written or oral test to assess theoretical understanding.

Practical Assessment the candidate performs specific trade tasks under supervision to demonstrate practical skills and competence. Evaluation the assessor observes and measures the candidates performance against the required standards.

Final result If the candidate meets all the competency requirements, they are declared competent and awarded a trade certificate this process ensures that only qualified and competent artisans are certified to work in their trade.

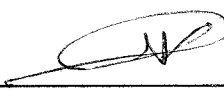
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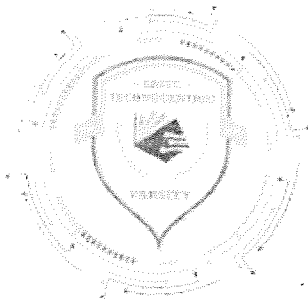
Assessor's Report

Candidate's Name	Xolani Malinga	ID No.	9511155424085
Assessor's Name		Reg. No.	
ASSESSMENT DECISION			
	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
Signature of Assessor	Signature of Candidate		

X.M

Candidate Feedback Report

Candidate's Name	Xolani Malinga		ID No.	9511155424085
Assessor's Name		Reg. No.		
Unit Standard Title				
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	



25
25
100%

INDUSTRY CONTEXT

FORMATIVE ASSESSMENT



Module #	671101000-KM-04:
NQF Level	level 3,
Notional hours	20
Credit(s)	Cr 2
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Contact Address	32467 Mandela Str Gx12
Telephone (H)	0747732317
Telephone (W)	
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language

Resources
Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Xolani Malinga

Learners Name: XOLANI COLEEN MALINGA

Learner's signature: 

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

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TO: The Internal Moderator

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Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

2014

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
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5	

Learner's signature: _____ Date: _____

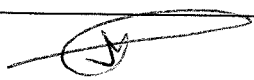
Employee witness: _____ Date: _____



PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Xolani Malinga</i>	Signature: 
Facilitator Name:	Signature:

[Faint signature]

Complete the following questions as per instructions provided

LEARNING UNIT 1:

KM-04-KT01: Introduction to the world of work and the electrical trade (100%)

Learning Outcome 1:	KT0101 The Electricians world of work
Learning Outcome 2:	KT0102 Career opportunities of qualified Electricians
Learning Outcome 3:	KT0103 The Electrician's responsibilities and duties
Learning Outcome 4:	KT0104 Legislation relating to apprentices in the Electrical trade
Learning Outcome 5:	KT0105 Trade test requirements

Task	Questions Description	Marks
1.	Explain the Legal aspects that includes the employment contract and roles and responsibilities of the employer and employee	10

The legal aspects of employment refer to the laws and agreements that regulate the relationship between the employer and the employee. These laws protect the rights of both parties and ensure fair and safe working conditions.

An employment contract is a legal agreement between the employer and the employee. It explains the terms and conditions of the job.

The contract includes

- 1) Job title and duties
- 2) Working hours
- 3) Salary or Wages
- 4) Leave entitlement

51 Notice period, for termination

a) Workplace rules and policies

Both the employer and employee must follow the terms stated in the contract

Roles and responsibilities of the employer
the employer has legal duties towards employees

e.g

✓ providing a safe and healthy working environment
paying employees fair wages on time

✓ providing the necessary tools and equipment for the job

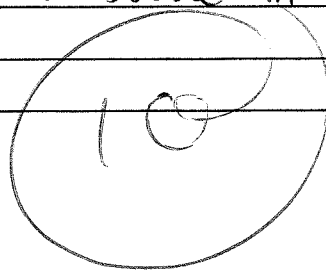
✓ treating employees fairly without discrimination
following labour laws and workplace regulations
providing training and supervision where required

Employers also have responsibilities in the workplace
these include

performing their work duties properly and honestly
following workplace rules and safety regulations
respecting supervisors and colleagues

taking care of company tools and equipment
being punctual and reliable

following the conditions stated in the employment contract



Task	Questions Description	Marks
2	Describe the use of applicable legislation relating to the electrical trade	5

In the electrical trade, legislation is used to ensure that all electrical work is done safely, legally and according to national standards. These laws protect electricians, workers, property and the public from electrical hazards such as shocks, fires and equipment damage.

One important law is the Occupational Health and Safety Act (OHSA). This act requires employers and employees to follow safe working procedures when working with electrical systems. It ensures that proper safety equipment, training and risk assessment are used to prevent accidents.

Another important regulation is the Electrical Installation Regulations under the OHSA. These regulations state that electrical installations must be done by qualified and registered electricians. They also require that a Certificate of Compliance (CoC) is used, issued after electrical work is completed to confirm that the installation meets safety standards.

11 (5)

Task	Questions Description	Marks
3	Explain Trade test methodology, requirements and assessment procedure leading up to the trade assessment	10

A trade test is a formal assessment used to determine whether a candidate has the required practical skills and theoretical knowledge to be certified as a qualified artisan in a specific trade.

The trade test methodology involves evaluating both practical and theoretical competence. Candidates are given tasks that simulate real workplace situations. The assessment is conducted by a qualified assessor at an accredited trade test centre. The candidate must demonstrate correct use of tools, safe working practices, problem-solving skills and the ability to complete tasks according to industry standards.

Before taking the trade test, the candidate must meet certain requirements. These usually include completing the required training or apprenticeship, gaining sufficient workplace experience and submitting a portfolio of evidence or logbook showing completed tasks.

The candidate must also be registered with the relevant authority and provide identification and proof of training.

The procedure leading up to the trade assessment includes applying for the trade test, verification of qualifications and experience and booking a test date at an accredited centre. During the assessment, the candidate performs practical tasks under supervision while the assessor

evaluates performance using set criteria. Safety, accuracy, quality of work and time management are all assessed. If the candidate meets the required standard, they are declared competent and awarded a trade certificate.

Training phase

the learner receives theoretical and practical training

Formative assessments

trainers evaluate the learners progress during training

Workplace experience

the learner gains hands on experience under supervision.

Preparation for trade test

the learner completes all required modules and logbooks

Final trade test assessment

the learner performs practical tasks and answers theory questions at an accredited test centre.

If the candidate passes, they receive a trade

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1				
2				
3				
4				

Learner Signature

Date

Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT

Learner's Total Mark:
not met ☐

Requirements met ☐

Requirements

Action/s required:

By when:

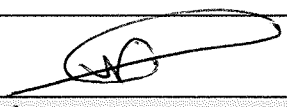
LEARNER FEEDBACK & COMMENTS

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

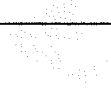
DECLARATION BY LEARNER

I, Xolani Colleen Malinga.....declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.

			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR

I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.

			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

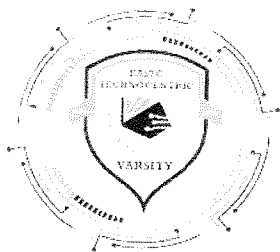
ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit



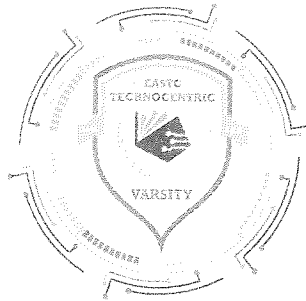
LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 05	Wireways and Wiring, NQF Level 4.	11

Xolani Malinga



38

AO

95.1%

WIRE WAYS AND WIRING

SUMMATIVE ASSESMENTS

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	XOLANI COLLEN
Surname	MALINGA
Cell NUMBER	0747732317
E-mail	collenx14@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed

Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

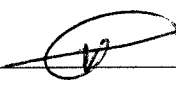
People to be involved with assessment

Learner: <u>Xolani Malinga.</u>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

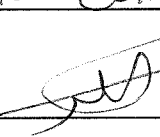
Guidelines to the learner:

Xolani Malinga 

Learners Name:

Xolani Colleen Malinga

Learner's signature:



Date: _____

Assessors Name: _____

Assessor's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments

- If not all evidence available was taken into account during the assessment
- The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
- The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
- Not all the range items were available for assessment

2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.



ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

LEARNING UNIT 1:

KM-05-KT01: Wire ways (35%)

Learning Outcome 1:	KT0101 Definition, types, purpose and applications of wireways
Learning Outcome 2:	KT0102 General provisions for wireways as prescribed in the SANS 10142-1

Task	Questions Description	Marks
1	a) Define the term wireway as given in the SANS 10142-1 Code of Practice	5
	b) Describe different types of wireways and their applications	5

A wire way is an enclosed channel or trough designed to contain and protect insulated conductors or cable allowing easy access along its length for inspection maintenance or replacement.

B) Metal Wireways

Made from metal for strength and durability

Application Used in industrial environments where mechanical protection is required and where they may be exposure to damage or harsh conditions

2. Non-Metallic Wireways

Made from plastic materials such as PVC

Applications Used in residential or commercial installations where corrosion resistance and electrical insulation are important.

11

10

3. Surface Mounted Wireways

Installed on the surface of walls or ceilings

Application

Used in situations where wiring need to be added without into walls

4. Flush Wireways

Installed inside walls or floors application
Used for neat, hidden wiring in new buildings.

5. Under Floor Wireways.

Installed beneath raised floors

Application

Common on offices for distributing power and data cables.

LEARNING UNIT 2:

KM-05-KT02: Wiring of installations (35%)

Learning Outcome 1:	KT0201 Regulations and statutory requirements for wiring of premises
Learning Outcome 2:	KT0202 Electrical diagrams and symbols
Learning Outcome 3:	KT0203 Electrical components and their applications

Task	Questions Description	Marks
2	List, identify and explain the meaning of all standard International Electrotechnical Commission (IEC) wiring symbols given on work drawings	10

Light Symbol = A circle with a cross inside

Meaning Indicates a lighting point or lamp

2. Switch (One-Way Switch) a break in a line with a small angled line.

Meaning Controls a light from one position

3. Two-Way Switch similar to one-way but with two connection points

Meaning Controls a light from two different positions

4. Socket outlets A circle with parallel lines inside

Meaning point where electrical appliances are connected

5. Distribution board A rectangle labelled (DB)

Meaning connection to earth for safety

6. Earth Symbol = three horizontal lines decreasing in size

Meaning connection to earth for safety

7. Junction box = Small circle or square

8. Conductor (Wire) = Straight line

9. Crossing wire = lines crossing without a dot

Meaning Wire cross but are not connected.

10. Connected Wire lines joined with a dot

Task	Questions Description	Marks
3	a) Describe the principles of operation of various control systems	5
	b) Describe the principles of operation of single and three phase circuit breakers and core balance earth leakage relays (wound primaries and straight primaries with tripping relay)	5

1. Input A Signal or command is given
2. Processing (Controller) the control device interprets the input decides what action to take
3. Output the system produces a response
4. feedback the output is monitored and fed back to the controller to maintain accuracy
5. Types of Control System
 - Open loop → No feedback
 - Closed loop → Uses feedback for accuracy

B. Single-Phase Circuit breaker

Protects a single-Phase Circuit

operation: When current exceeds the rated value, a thermal or magnetic mechanism trips and opens the circuit

2. Three-Phase Circuit breaker

Protects three-Phase Systems

operation: If a fault occurs in any phase, the breaker trips and disconnects all three phases simultaneously

LEARNING UNIT 3:

KM-05-KT03: Earthing and bonding (30%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Describe the earthing systems relevant to SANS 10142-1.	3
	b) Describe how a common earth electrode is used in reticulation circuits	3
	c) Describe the provision of earthing for underground cables and overhead lines	4

TN System (Terra Neutral)

Supply neutral is directly earthed

Exposed conductive parts connected to supply earth

Variants

TN-S: Separate PE (earth) and N conductors throughout.

TN-C: Combined PEN conductor (Neutral + earth).

TN-C-S (PME): Combined PEN in part of system, separated at installation.

TT System (Terra Terra)

Supply neutral earthed at source

Installation has its own independent earth electrode

No direct connection between supply earth and installation earth

10

3. Core balance earth leakage relay

Principle Detects imbalance between live and neutral currents

Under normal conditions

Currents in live - Current in neutral No leakage

Under Fault

Some current flows to earth Imbalance occurs = relays trips.

Types:

a. Wound primary type
Conductors are wound

More sensitive and Used in smaller systems.

b. Straight Primary type

Conductors pass straight through the core.
common in practical installations

With tripping relay when leakage is detected the relay sends a signal to trip the circuit breaker, disconnecting the supply to prevent shock or fire.

b) A common earth electrode is shared by multiple points in a distribution (reticulation) network.

Typical in TN-C-S / PME systems.

The neutral is earthed at multiple points along network.
All consumers connect to the same earthing system via the supply PEN conductor.

Functions: Maintains equipotential bonding across the network.

Provides low impedance fault path for fast fault clearance.
Stabilizes system voltage relative to earth.

Important considerations

Adequate conductor sizing (PEN).

Multiple earthing points to reduce earth resistance.

Proper bonding to avoid dangerous touch voltages.

c) Metallic sheath/armour must be earthed at one or both ends.

Provides:

Fault current path.

Protection against insulation failure.

Methods:

Single-Point bonding (reduces circulating currents).

Both-end bonding (improves fault clearing).

Joint boxes and terminations must be bonded to earth.

Cable screens connected to earth to control induced voltages.

Overhead lines

Poles/towers are earthed to dissipate lightning and fault currents.

Use of:

Earth wire (shield wire) above conductors for lightning protection

Pole earthing electrodes (rods or plates)

Stay wire must be earthed or insulated.

Neutral conductor often earthed at multiple especially in TM systems.

Assessor's Report

Candidate's Name	Xolani Cotten Mdingo		ID No.	9511155924085
Assessor's Name			Reg. No.	
Title	Wire Ways and Wiring.			
ASSESSMENT DECISION				
Outcome	C	NYC	Comments	
OVERALL COMPETENCE JUDGEMENT:				
Comments				
Date				
Signature of Assessor	Signature of Candidate			

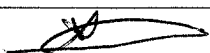
Candidate Feedback Report

Candidate's Name	Xolani Collen Malinga		ID No.	9511155429085
Assessor's Name		Reg. No.		
Title				
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good.	
Overall Assessment Decision				
Additional Notes				
Date				
Signature of Assessor			Signature of Candidate	

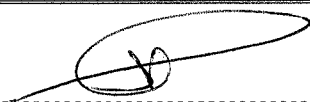
Candidate Appeal Form

Candidate's Name:		ID No.	
Assessor's Name:		Reg. No.	
Title:			
Date:			
SECTION 1			
Candidate's reason for disagreeing with the assessment decision			
Assessor's rationale for the assessment decision			
Candidate's signature			
Assessor's signature			
SECTION 2			
Moderator's decision			
Moderator's signature			

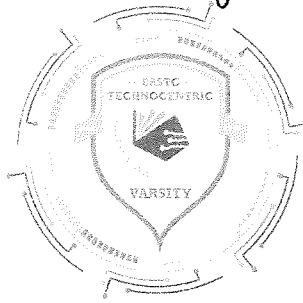
Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Xolani C. Malinga	ID No.	981155424085
Title	Wire Ways and Wirings		
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Xolani Malinga</i>	NAME of ASSESSOR		
VENUE <i>EastC University</i>	DATE of REVIEW		
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
			
Learner	Date	Assessor	Date
		Moderator	Date

Xolani Malinga



29

30

97%

Wire ways and wiring, FORMATIVE ASSESSMENT

Module #	671101000-KM-05:
NQF Level	level 4,
Notional hours	110
Credit(s)	Cr 11
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Xolani COLLEN Malinga
Contact Address	32461 Mandela Str Ex12B
Telephone (H)	0747732317
Telephone (W)	0747732317
Facsimile	N/A
Cellular	0747732317
E-mail	collenx14@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓
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Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources

Barriers

People to be involved with assessment

Learner: <u>Xolani Malinga</u>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

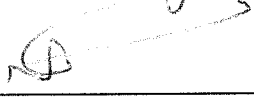
Next steps for learning

Resources required for this assessment

Guidelines to the learner:

Xolani Malinga

Learners Name: Xolani Colleen Malinga

Learner's signature: 

Date:

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances

- If the laid down assessment procedures were not followed during assessments
- If not all evidence available was taken into account during the assessment
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- The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
- Not all the range items were available for assessment

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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____

10/10/2020

Complete the following questions as per instructions provided



LEARNING UNIT 1:

KM-05-KT01: Wire ways (35%)

Learning Outcome 1:	KT0101 Definition, types, purpose and applications of wireways
Learning Outcome 2:	KT0102 General provisions for wireways as prescribed in the SANS 10142-1

Task	Questions Description	Marks
1.	a) Describe and explain the provisions in the SANS 10142-1 Code of Practice for the correct and safe installation and use of the various types of wireways	5
	b) Describe factors that influences the selection of wireways	5

1. Selection and Installation: Wireways must be suitable for the environment (mechanical protection, temperature, chemical exposure) and comply with the required IP rating.
2. Mechanical protection they must protect cables from damage and prevent ingress of foreign objects or moisture.
3. Support and Fixing proper spacing of supports and secure fixing are required to avoid sagging or stress on cables.
4. Bending Radii Minimum bending radii for cables in wireways must be observed to prevent damage to insulation.
5. Accessibility Wireways should allow easy access for inspection, maintenance and future modifications.
6. Earthing and bonding Metallic wireways need proper earthing to ensure electrical safety and prevent shock hazards.
7. Load and Capacity the wireways capacity must match the number and size of cables it will contain, respecting derating factors.

Environment conditions: temperature, moisture, chemicals exposure
fire risks or UV exposure.

2. Type of cable: Size, number, insulation, type and whether they require shielding.

Mechanical requirements: need for protection against impact vibration or physical damage.

3. Installation method: Surface-mounted, embedded or concealed systems.

4. Maintenance needs ease of access for future additions or repairs.

Cost and aesthetics material choice (metal plastic) and appearance in the in the installation

5. Regulatory compliance: adherence to SANS 10142-1 and other local codes for safety and performance.

Maintenance needs: ease of access for future additions or repairs.

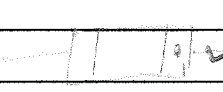
LEARNING UNIT 2:

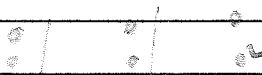
KM-05-KT02: Wiring of installations (35%)

Learning Outcome 1:	KT0201 Regulations and statutory requirements for wiring of premises
Learning Outcome 2:	KT0202 Electrical diagrams and symbols
Learning Outcome 3:	KT0203 Electrical components and their applications

Task	Questions Description	Marks
2	a) Identify electrical components and draw schematic diagrams of installations	3
	b) State and explain the safety purpose of earthing, fuse, circuit breakers and earth leakage protection unit	2

a) To address this you need to list typical electrical components used in installations (e.g. switches, transformers, cables)

Fuse  Fast blow fuse symbol.

Contactor 

5

b) Earthing provides a safe path for fault currents to the ground preventing electric shock and equipment damage by keeping the systems at zero.

Fuse: A protective device that melts and breaks the circuit when excessive current flows

Circuit breaker: An automatic switch that trips and opens the circuit during overcurrent

Earth leakage Protection Unit (ELPU) RCD: Detects small leakage currents to earth, tripping the circuit to prevent electrocution and fire risks.

Task	Questions Description	Marks
3	Describe the purpose of load distribution, lightning arrestors and energy control units.	5

1. load distribution

To evenly spread electrical load across circuits or phases
 prevents overloading of any single circuit
 Improves system efficiency
 ensures stable Voltage supply

lightning arrestors

to protect electrical systems and equipment from lightning surges
 Protection against damage to electrical equipment
 Fire hazard
 Insulation breakdown.

3. Energy Control Units

to ~~main~~ monitor, control and optimize electrical energy usage
 reduce ~~overload~~ electricity costs
 prevents overload conditions
 Improves energy efficiency

load distribution ensures the system operates efficiently

lightning arrestors protect against external high-voltage surges

Energy control Units improve energy management and excessive demand.

LEARNING UNIT 3:

KM-05-KT03: Earthing and bonding (30%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation	2
	b) Explain the term 'systems earthing'	3

Earthing on the Supplier Side

* Neutral-earth connection at source the transformer neutral must be connected to a solid earth electrode.

ensure a defined reference point for the system Voltage earthing on the consumer side

* Neutral-earth bonding at main distribution board (MDB)

the neutral conductor should be connected to the main earth electrode inside the consumer installation.

B. System earthing is the process of connecting one point of an electrical system to earth in a controlled manner to stabilize voltage during normal operation facilitate the operation of protective device during faults

Task	Questions Description	Marks
4	Explain the various processes of measuring, testing and calculating earthing and bonding system values.	5

1. Measuring Earth resistance

a) Fall of potential method (three-point test)
place the earth electrode^(e), a current probe^(c) and potential probe^(p) in a straight line on the ground.
pass a known current from E to C

$[R = \frac{V}{I}]$ calculate resistance Using ohm's law.

2. Testing bonding and Continuity

a) Continuity test
- Disconnect the circuit from the supply
Measure resistance between metal parts and earth bar
resistance should be very low

3. Calculating Earthing Values

a) Resistance of single Rod earth electrode
 $R = \rho \times \frac{L}{A}$ (factor)

b) Resistance of multiple electrode in parallel
 $R_{total} = \frac{R_i}{n \text{ number of electrodes}}$

c) Fault Current Calculation = $\frac{V}{R_e}$

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

Outcomes		1	2	3
1	KT0201 Regulations and statutory requirements for wiring of premises			✓
2	KT0202 Electrical diagrams and symbols			✓
3	KT0303 Regulatory and statutory requirements related to earthing and bonding			✓



Learner Signature

Date



Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

Good work

ASSESSMENT JUDGEMENT

Learner's Total Mark:

Requirements met ☒

Requirements not met ☐

Action/s required:

By when:

LEARNER FEEDBACK & COMMENTS

Very good Mr Malinga.

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER	
I, <u>Xolani Colleen Malinga</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.	
	
Learner	Date

DECLARATION BY THE ASSESSOR	
I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.	
	
Assessor	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

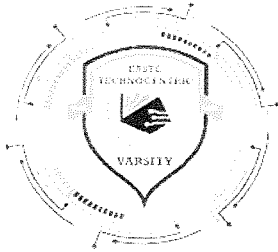
ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit



LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 06	Rotating Electrical Machinery, NQF Level 4.	13

Colon: Malinga

59
69
867

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

**ROTATING ELECTRICAL MACHINERY
FORMATIVE ASSESSMENT**

Details	Please Complete this Section
Name	XOLANI COLLEN MALINGA
Surname	Malinga
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	Mbali

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Xolani Malinga</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner:

<i>Xolani Malinga</i>

Learners Name: *XOLANI COLLEN MALINGA*

Learner's signature:  Date: _____

Facilitator Name: _____

Facilitator's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

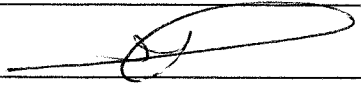
Employee witness: _____

Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

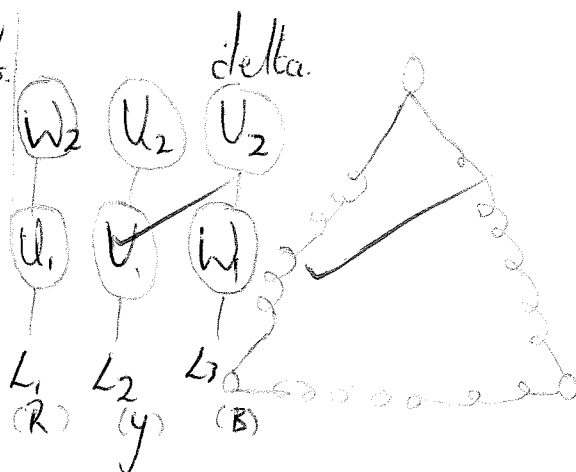
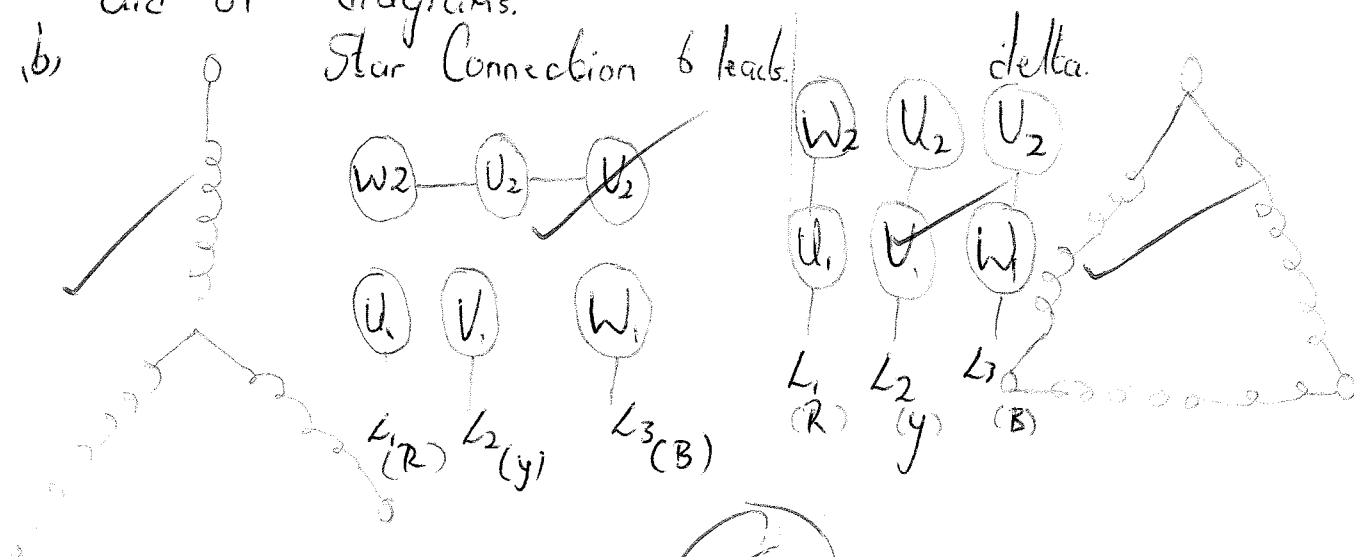
Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <u>XOLANI COLLEEN MALINGA</u>	Signature: 
Facilitator Name:	Signature:

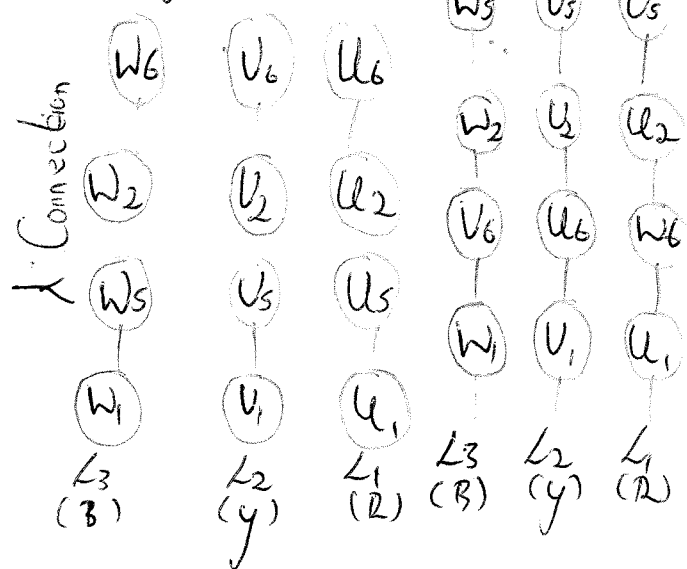


Complete the following questions as per instructions provided

b) aid of diagrams.
Star Connection 6 leads.

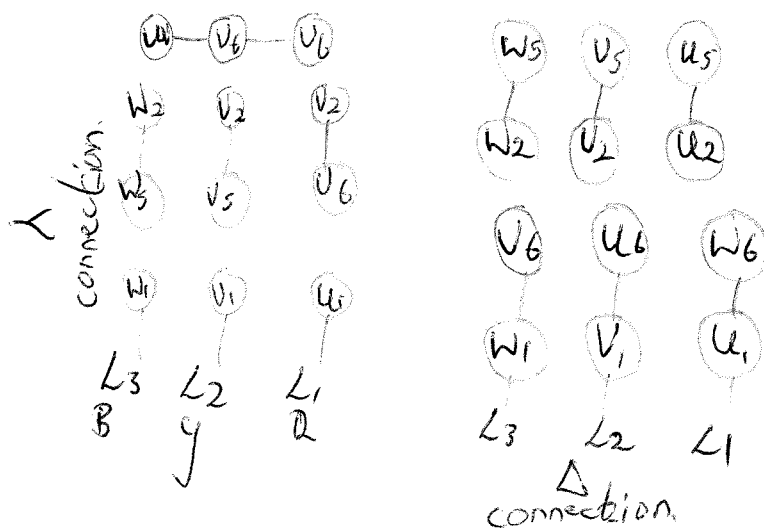


Single Voltage running and Δ connection.

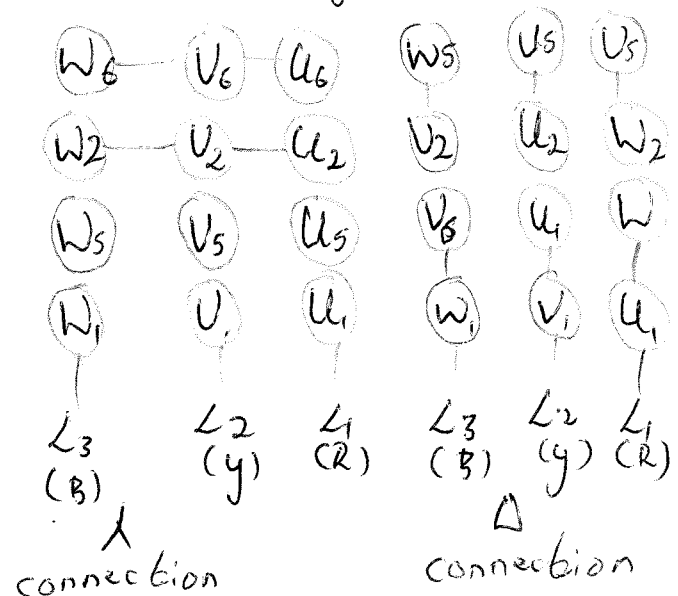


Dual Voltage Δ Starting and Δ starting (12 leads)

High Voltage Connection



Low Voltage Connection.



LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction	5
	b) Describe with the aid of diagrams, the configuration of motor connections	5
	c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	10

Single-Phase induction motor

Main parts: stator, rotor, shaft, bearings, frame, cooling fan
 Construction the stator has a single-phase winding, the rotor is usually a squirrel-cage rotor.

Common starting parts: starting winding, capacitor, centrifugal switch or shaded pole ring

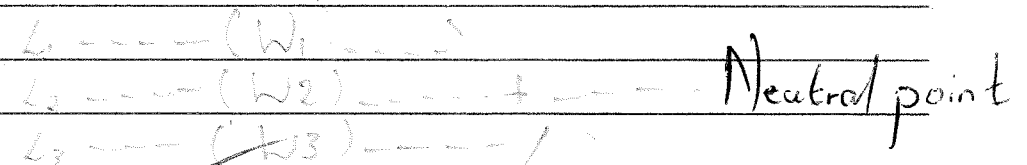
Three-phase induction motor

Main parts, stator, shaft, bearings, frame.
 Construction the stator has three windings placed 120° apart electrically. The rotor is usually either
 • Squirrel-cage rotor
 • Wound rotor

Configuration of motor connections

Star (Y) connection

In a star connection, one end of each winding is joined together at a common point

- Lower phase V
 - Used for starting large motor
 - Reduces starting current
- 

Delta (Δ) connection

In a delta connection, the windings are connected end-to-end in a closed loop.

(W1)

L1 --- (W1) --- L2

(W2) (W3)

Full line voltage across each winding
higher torque

Common in running condition of motors.

Influence of time delay and current rating

A time delay allows the motor to start without the overload device tripping immediately

Motors draw a high inrush current at startup. The overload device must be set close to the motor's full load current (FLC).

A locked rotor causes very high current for a long time.

Overload relays do not mainly protect against short circuit.

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of alternating current motors	4
	b) Describe and explain how tests are conducted on alternating current motors	4
	c) Describe and explain protection devices used for alternating current motors	5
	d) Compare the advantages and disadvantages of single- and three-phase motors	6

(a) the principle of alternating Current When an alternating current flows through the stator windings it produces a rotating magnetic field (RMF). This rotating magnetic field moves around the stator at synchronous speed. The stator is connected to the AC supply and produces a

magnetic field.

this Rotating magnetic field cuts the rotor conductors

According to Faraday's law of Electromagnetic Induction a voltage is induced in the rotor

The induced Voltage causes current to flow in the rotor conductors

b) Tests Conducted on alternating current motors the motor is running without mechanical load.

Measures input power, current and Voltage

Used to determine core losses, friction losses and magnetizing current

c) Blocked rotor test (Short Circuit test)

the rotor is locked so it cannot rotate

A reduced Voltage is applied

d) Insulation resistance test

Done using a megger

Checks the insulation quality between windings and earth.

Temperature rise test

the motor runs under load for long time

ensure the motor temperature remains within safe limits

load test

Motor is operated with different loads

Advantage and disadvantages of Single-Phase and three

Power Supply Uses single-phase supply Uses three phase supply

Starting Not Self-Starting

Efficiency low efficiency

LEARNING UNIT 2:**KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)**

Learning Outcome 1:	KT0201	Construction of direct current motors
Learning Outcome 2:	KT0202	Principles of operation for direct current motors
Learning Outcome 3:	KT0203	Configuration of direct current motor connections
Learning Outcome 4:	KT0204	Types and application of direct current motors
Learning Outcome 5:	KT0205	Testing principles of direct current motors
Learning Outcome 6:	KT0206	Protection of direct current motors
Learning Outcome 7:	KT0207	Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) List the different types of direct current motors and describe their construction	2
	b) Describe the applications of different types of direct current motors	2
	c) Describe with the aid of diagrams, the configuration of motor connections, power circuits and control circuits.	2
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation, short circuits.	2
	e) Calculate direct current motor properties	2

(a) Types of direct Current motors and their construction

1. DC Shunt Motor
2. DC Series Motor
3. DC Compound Motor
4. Permanent Magnet DC Motor (PMDC)

Construction of DC motor, DC motor consist of the following

Stator (Field System)

The stationary part of the motor produces the magnetic field using field windings or permanent magnets.

Rotor (Armature)

The rotating part of the motor.

Contains armature windings where current flows.

Commutator

A cylinder/copper segment connected to the armature.

Converts alternating current in the armature into direct current.

Brushes

Made of Carbon

Provide electrical contact between the power supply and the rotating commutator.

Shaft and bearings

Support rotation and transmit mechanical power.

(b) Application of different types of DC motors.

DC Series motor

Produces very high starting torque.

Applications

Electric trains

Cranes

Hoists

Elevators

DC Shunt Motor

Runs at constant speed

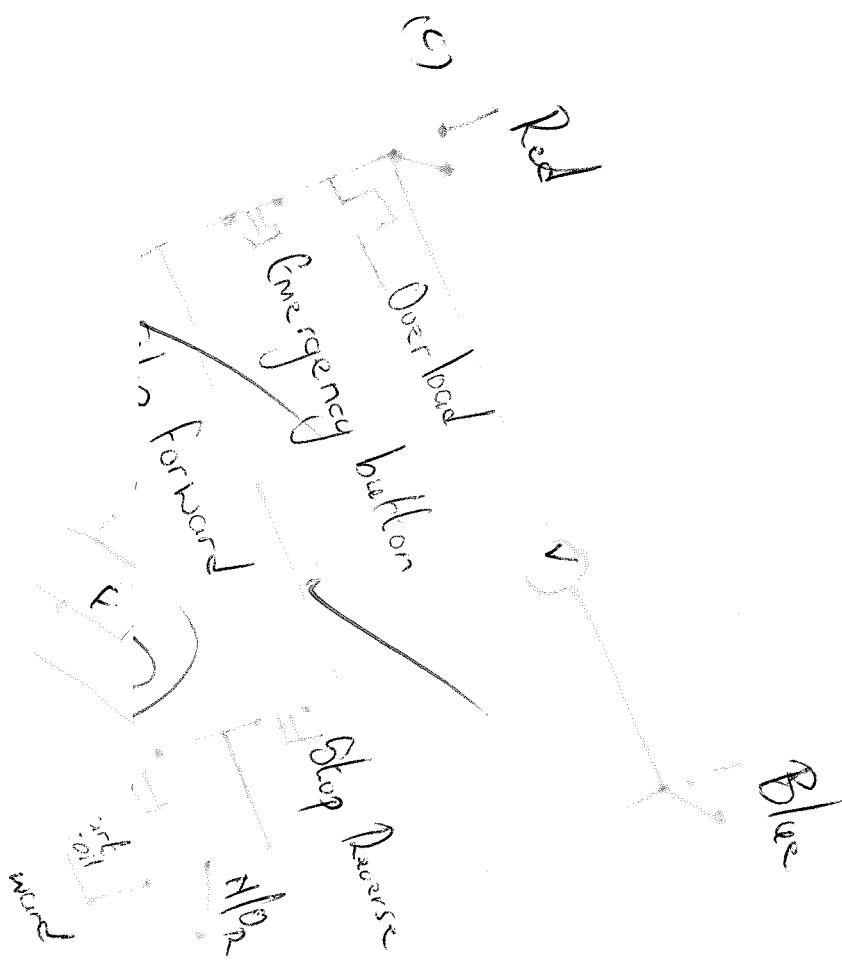
Lathes

Pump

Fans

Machine tools

Conveyors



(d)
DC Compound Motor

Combines features of Series and Shunt Motors

Rolling mills

Press machine

Ho. Head planers

Elevators

Permanent Magnet DC

Small size and Simple Construction

toys

Windshield wipers

Portable tools

Computer devices

(e) Configuration of Motor connections, power circuits and Control Circuits

Series connection Field winding connected in series with armature

Shunt connection Field winding connected parallel to armature

$$E_b = V - I_a R_a$$

$$P = V \times I$$

P = Power (Watts)

V = Voltage

I = Current

LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
3	a) Explain the principle of operation of alternators and generators	6
	b) Describe the applications of different types of alternators and generators	4
	c) Describe and explain protection devices used for alternators and generators	4
	d) Explain how time delay and current rating of overload protection devices influence their use in protecting alternators and generators from damage in the case of locked rotors, overload during operation and short circuits.	6

(a) Both alternators and generators convert mechanical energy into electrical energy using the principle of electromagnetic induction.

The principle follows Faraday's law of Electromagnetic Induction, which states that when a conductor moves through

a magnetic field, an electromotive force (EMF) is induced in the conductor. (E)

A prime mover (engine, turbine or motor) rotates the rotor. The rotor produces a magnetic field. As the rotor spins, the magnetic field cuts across the stator windings.

(b) Difference.

Alternator produces alternating current (AC). generator (DC generator) produces direct current (DC) using a commutator. (F)

Applications of different types of Alternators and generators. Used in vehicles to charge the battery, supply electrical power to lights, ignition system, radio.

(c) Protection devices used for alternators and generators use fuses or circuit breakers.

Protects against excessive current by faults.

How time delay and Current rating influence overload protection.

If the current exceeds the rated values the protection device will trip or disconnect the circuit. This prevents overheating and damage to windings. (H)

Time delay allows temporary current increases without immediately tripping. (I)

Does not trip unnecessarily.

But disconnects the system if the overload continues too long.

c) Protection in different fault cases

~~Locked rotor~~
Current rises very high
protection trips after a short delay to avoid
~~overheating~~

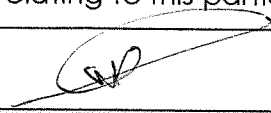
Overload during Operation
Current slightly above normal
Device trips after longer delay

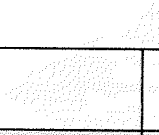
Short Circuit
Very high Current
protection trips instantly to prevent severe damage

Reverse Power protection
prevents the generator from running as a motor
if the prime mover fails

Earth fault protection
Detects leakage current to earth

Over/Under Voltage protection
protects equipment from dangerous voltage levels

DECLARATION BY LEARNER			
I, <u>Xolani Collen Malinga</u> declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments 4. Conduct observation in workplace where applicable

5. Complete learner toolkit

6. Complete assessor toolkit



34
40
85%

Module #	671101000-KM-06:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

**ROTATING ELECTRICAL MACHINERY
SUMMATIVE ASSESMENT**

Details	Please Complete this Section
Name & Surname:	Xolani Colen Malinga
Organisation:	EASTC TECHNOCENTRIC VARSITY
Unit/Dept:	ELECTRICAL ENGINEERING
Facilitator Name:	Mbali

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
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	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	☒
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	☒
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	☒
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	☒
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	☒
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	☒
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	☒
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	☒
9	Ensure the assessment environment is appropriate or make special arrangements	☒
10	Discuss moderation	☒
11	Allow the learner opportunity to clarify any items discussed	☒

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>XOLANI COLLEEN Malinga</i>	Signature <i>[Signature]</i>
Assessors Name:	Signature:



INSTRUCTIONS:

The following sections will test your knowledge, and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-06-KT01: Rotating electrical machinery - AC motors (35%)

Learning Outcome 1:	KT0101 Construction of alternating current motors
Learning Outcome 2:	KT0102 Principle of operation of alternating current motors
Learning Outcome 3:	KT0103 Configuration of motor connections
Learning Outcome 4:	KT0104 Types of single phase and three phase alternating current motors
Learning Outcome 5:	KT0105 Application of alternating current motors
Learning Outcome 6:	KT0106 Testing principles of single phase and three phase alternating current motors
Learning Outcome 7:	KT0107 Protection of motors
Learning Outcome 8:	KT0108 Calculation of motor properties

Task	Questions Description	Marks
1	a) List the different types of alternating current motors and describe their construction	5
	b) Describe with the aid of diagrams, the configuration of motor connections	5
	c) Explain how time delay and current rating of overload protection devices influence their use in protecting motors from damage in case of locked rotors, overload during operation and short circuits.	10

Different types of Alternating current motors and their construction.

Stator: Contains three-phase windings connected to the AC supply which create a rotating magnetic field.

Rotor: Usually has DC excitation windings or permanent magnets.

Slip rings and brushes (In some designs) supply DC to the rotor.

Shaft and bearings allow rotation.

The rotor locks with the rotating magnetic field and rotates at constant synchronous speed.

Induction Motor (Asynchronous Motor)

Stator: laminated iron core with three-phase windings connected AC Supply.

Rotor: Either squirrel cage rotor or wound rotor.

Air gap: Small gap between stator and rotor
Shaft, bearings and frame.

Types of Induction motors:

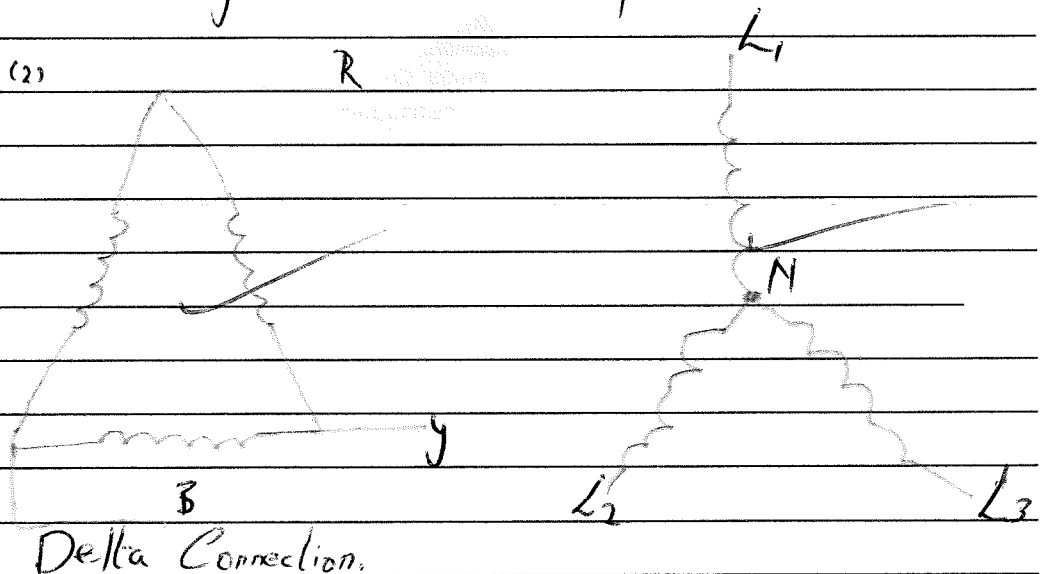
Squirrel cage induction motor

Slip ring (wound rotor) induction motor.

Single-Phase AC Motor.

Single Phase motors operate on single-Phase AC Supply and are commonly used in small appliances.

b. Diagram (2)



LEARNING UNIT 2:

KM-06-KT02: Rotating Electrical Machinery - DC motors (35%)

Learning Outcome 1:	KT0201 Construction of direct current motors
Learning Outcome 2:	KT0202 Principles of operation for direct current motors
Learning Outcome 3:	KT0203 Configuration of direct current motor connections
Learning Outcome 4:	KT0204 Types and application of direct current motors
Learning Outcome 5:	KT0205 Testing principles of direct current motors
Learning Outcome 6:	KT0206 Protection of direct current motors
Learning Outcome 7:	KT0207 Calculation of direct current motor properties

Task	Questions Description	Marks
2	a) Explain the principle of operation of direct current motors	2
	b) Describe and explain how various tests are conducted on direct current motors	2
	c) Describe and explain protection devices used for direct current motors	3
	d) Compare the advantages and disadvantages of different types of direct current motors	3

a) A direct Current (DC) motor works on the principle that a current carrying conductor placed in a magnetic field experiences a mechanical force.

When DC Supply is applied to the armature windings current flows through the conductors.

The armature conductors are located inside the magnetic field produced by the field windings

b, Insulation resistance test

Concluded Using a megger

Measures the insulation resistance between windings and the motor frame.

Ensures that insulation is in good condition and there are no leakage currents

(2) No load test

The motor is running without any mechanical load. Voltage, Current and Speed are measured.

Used to check the general operation and losses of the motor.

Overload protection

Uses overload relays or circuit breakers.

Disconnects the motor when current exceeds the rated value

Prevents Overheating

Short Circuit protection

Fuses or circuit breakers are used

They quickly disconnect the supply when a short circuit occurs

Under-Voltage protection

Protects the motor when the supply voltage drops too low

Prevents damage or improper operation of the motor

Types DC motor

Series motor

Shunt motor

Compound motor

Advantages

Very high starting torque

good speed regulation

(2) Combines good torque and speed regulation.

LEARNING UNIT 3:

Rotating electrical machinery - Alternators and Generators (30%)

Learning Outcome 1:	KT0301 Construction of alternators and generators
Learning Outcome 2:	KT0302 Principle of operation of alternators and generators
Learning Outcome 3:	KT0303 Configuration of alternator and generator connections
Learning Outcome 4:	KT0304 Types and application of alternators and generators
Learning Outcome 5:	KT0305 Testing principles of alternators and generators
Learning Outcome 6:	KT0306 Protection of alternators and generators
Learning Outcome 7:	KT0307 Calculate properties of alternators and generators

Task	Questions Description	Marks
2	a) List the different types of alternators and generators and describe their construction	2
	b) Explain the fundamental differences between alternators and generators	2
	c) Describe and explain how tests are conducted on alternators and generators	3
	d) Describe with the aid of diagrams, the configuration of alternator and generator connections, power circuits and control circuits.	3

Alternator
Consists of a Stator and rotor
The stator contains armature windings where AC voltage is generated.
The rotor carries the field winding and produces a rotating magnetic field.

DC generator.

Consists of a field system, armature, commutator and brushes

The armature rotates inside the magnetic field
The commutator converts the generated AC into DC output.

b) Alternator

produces alternating current (AC)

Uses slip rings

Output current changes direction periodically

DC generator

produces direct current (DC)

Uses commutator

Output current flows in one direction

Insulation resistance test

Done using a megger

Checks insulation between windings and the machine frame.

Open Circuit test

Alternator is run at rated speed with no load
Output voltage is measured to check performance

load test

generator or alternator is connected to a load
Voltage, current and efficiency are measured during operation.

power circuit

Carries main electrical power from the generator or alternator to the load

Includes components such as Switches, Circuit breakers and Cables

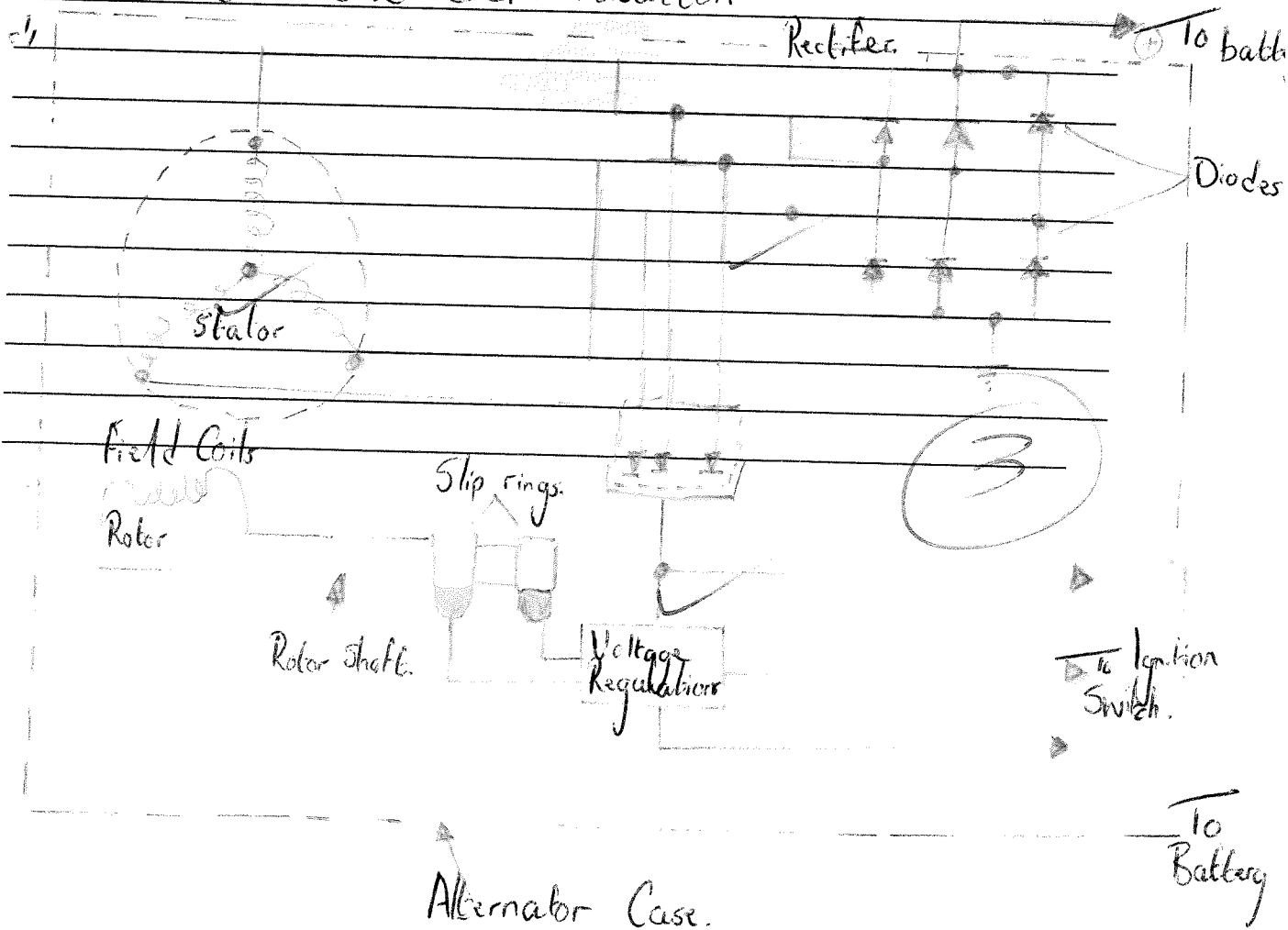
Control Circuits

Used to control and protect the machine operation
Includes relays, contactors, push buttons and protection devices

Connection Configuration

Alternators are commonly connected in star (Y) or delta (Δ) for three phase power systems

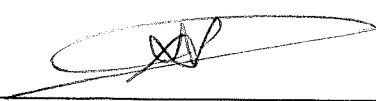
generators supply power through armature windings connected to external circuits brushless and commutator



Assessor's Report

Candidate's Name	<i>Xolani Collen Malinga</i>	ID No.	9511155424085.
Assessor's Name		Reg. No.	
ASSESSMENT DECISION			
	C	NYC	Comments
KT0101 Construction of alternating current motors	✓		
KT0302 Principle of operation of alternators and generators	✓		
KT0303 Configuration of alternator and generator connections	✓		
KT0304 Types and application of alternators and generators	✓		
KT0305 Testing principles of alternators and generators	✓		
KT0306 Protection of alternators and generators	✓		
KT0307 Calculate properties of alternators and generators	✓		
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
Signature of Assessor			
		Signature of Candidate	

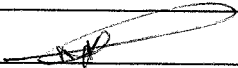
Candidate Feedback Report

Candidate's Name	Xolani Collen Malinga		ID No.	9511155424085.
Assessor's Name			Reg. No.	
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Xolani Collen Malinga.	ID No.	9511165424085.
MODERATION DECISION			
	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Volani College Malinga</i>		NAME of ASSESSOR	
VENUE <i>Coast Technocentre</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
Moderator		Date	

